

SAI VIVEKANAND REDDY

☎ 857-465-9966 | ✉ vangala.sa@northeastern.edu | 🌐 in/vangalas | 🐙 sai-vivekanand | 🌐 sai-vivekanand.github.io

Education

Northeastern University, Boston, USA

September 2023 - December 2025

Master of Science, Computer Software Engineering

- **Relevant Courses:** Data Science Engineering, MLOps, Networks & Cloud Computing, Intelligence Analytics

BITS Pilani, Pilani, India

August 2016 - June 2020

Bachelor of Engineering, Electronics Engineering

- **Relevant Courses:** OOPs, AI, DBMS, Algorithms, Operating Systems, Neural Networks, Distributed Systems

Technical Skills

- **Languages:** Python, Java, Go (Systems & Concurrency), SQL, TypeScript/JS, HTML5/CSS
- **Cloud & DevOps:** GCP, AWS (S3, EC2, Lambda, RDS), Docker, Kubernetes, Terraform, CI/CD, Packer
- **AI/ML:** Gen AI, LangChain, LangGraph, NLP, PyTorch, TensorFlow, scikit-learn
- **Frameworks & Tools:** Spring Boot, FastAPI, React, Git, Spark, Airflow, MySQL, PostgreSQL, Redis

Work Experience

Humanitarians AI

Boston, USA

AI Engineer

Jan 2025 - May 2025

- Achieved over **95% accuracy** in multi-modal document processing by engineering a content analysis engine with the Gemini Vision API to handle 10+ file formats, including image-based table extraction.
- Architected a 4-tier **persistent memory system** for AI agents, enabling semantic search and long-term knowledge retention by integrating **ChromaDB** for vector storage and **OpenSearch** for immutable, large-scale audit logging.
- Deployed highly reliable AI services with **99.9% uptime** by orchestrating a complete **CI/CD pipeline** using Terraform, Docker, and Kubernetes for automated, zero-downtime deployments.

Puddl

Bangalore, India

Member of Technical Staff

June 2022 - August 2023

- Engineered a client-side analytics platform for OpenAI API monitoring, ensuring **100% data privacy** by architecting a **React and TypeScript** Single Page Application (SPA) that processed sensitive API keys exclusively in the browser.
- Orchestrated **CI/CD pipelines** to deploy VMs with load balancers and auto-scaling using Docker, Kubernetes, and Terraform, achieving **99.9% service uptime** and high reliability.

Blue Yonder

Hyderabad, India

Software Engineer

July 2020 - May 2022

- Reduced initial load times by **40%** for data-heavy users by re-architecting the data-loading mechanism, implementing dynamic SQL query construction in Java to selectively fetch only user-requested measures.
- Enhanced user productivity by **25%** by re-engineering a monolithic 'Save & Calc' feature into discrete actions, leveraging server-side HTTP session caching to enable rapid, database-free "what-if" forecast simulations.
- Diagnosed and resolved over **70+ customer-reported bugs annually** by debugging a full-stack Java Spring MVC application and implementing robust fixes with comprehensive **JUnit** tests.

Projects

Cloud-Native Web App (Java, Spring Boot, Terraform)

- Engineered a scalable, 3-tier web application on GCP, achieving high availability by deploying a **Spring Boot API** to an **auto-scaling instance** group and a private Cloud SQL DB.
- Automated **infrastructure provisioning** for 15+ GCP resources (VPC, KMS, LB, SQL), reducing manual deployment time by **100%** using Terraform and Packer for immutable, version-controlled infrastructure.
- Designed an **event-driven, asynchronous** email verification system using **GCP Pub/Sub** and Python **Cloud Functions**, decoupling the service from the main Spring Boot application to improve resilience.

Multi-Agent RAG Research System (Python, LangGraph, Airflow)

- Architected a **multi-agent research system** that synthesizes information from parallel web, academic, and Arxiv AI agents by orchestrating the workflow using Python, LangGraph, and a FastAPI backend.
- Automated a **multi-modal** document ingestion pipeline using Apache **Airflow** to process and index diverse file formats (PDFs, images) into a Pinecone vector database for efficient semantic retrieval.

Distributed Deep Learning Training Pipeline (PyTorch, HPC)

- Achieved a **3.7x speedup** on a U-Net text-to-image model by designing and implementing a multi-GPU parallel processing pipeline using PyTorch Distributed Data Parallel (DDP) on an HPC cluster.
- Reduced training time by an additional **66%** by integrating Automatic Mixed Precision (AMP), optimizing hardware utilization and memory efficiency.